

Dense paddock-scale validation of statistical and process-based soil-moisture downscaling

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Companion validation report to “Downscaling SMIPS soil moisture to 30 m: statistical and process-based estimation of root-zone soil moisture, with validation in the Murrumbidgee catchment”. This draft scaffold separates the model-development evidence from the dense-site validation evidence. Red [TODO: items] should be resolved before circulation.

Abstract

[TODO: Write after results are stabilised. Suggested structure: one sentence on the paddock-scale validation gap; one sentence describing Esdale, Tarrawarra and Llara; one sentence on the model-agnostic validation table; one sentence with headline model6/model8 skill and seasonal/wetness bias; one sentence on local calibration/spiking; one sentence on implications for deployment.]

Root-zone soil-moisture products are increasingly evaluated against sparse station networks, but management decisions occur at sub-property scale where within-paddock redistribution by terrain, soil and antecedent wetness is critical. Here we use three dense point datasets to test whether the final statistical and process-based models developed in the companion paper preserve paddock-scale soil-moisture patterns when treated as black-box gridded predictors. Skill is summarised using Pearson correlation (r), Nash–Sutcliffe efficiency (NSE), RMSE, ubRMSE and bias so that temporal association, error magnitude and mean-level offsets can be separated. [TODO: Insert headline values and conclusion.]

Keywords: soil moisture; dense validation; downscaling; paddock scale; model evaluation; local calibration; Nash–Sutcliffe efficiency

1 Introduction

Operational soil-moisture products generally resolve kilometres rather than management units, yet irrigation, trafficability, grazing, crop establishment and ecophysiological modelling often require information at the scale of a paddock or sub-property unit. The companion paper develops two routes to 30 m root-zone soil moisture: a statistical downscaler of the ≈ 1 km **SMIPS TotalBucket** product using terrain, soil and meteorological features, and a process-based bucket model with a static soil–terrain–climate readout. Its central validation question is whether these models transfer across stations, years and spatial blocks in the OzNet monitoring network (Smith et al., 2012).

That station-network validation is necessary but not sufficient for the intended product. Sparse stations can test regional transfer and temporal dynamics, but they cannot directly test whether a 30 m map reproduces within-site moisture redistribution. Dense point datasets provide a “microscope” over the downscaling field: they can reveal whether residuals cluster by terrain position, wetness state or season, and whether competing models fail in the same spatial locations.

This report therefore asks three questions:

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1. **Independent paddock-scale transfer.** When evaluated at dense point observations not used for model development, how much skill do model 6 RF and model 8 process retain?
2. **Seasonal, wetness-state and terrain-conditioned bias.** Are poor predictions concentrated in particular seasons, dry/wet states or model-input terrain/soil strata?
3. **Deployability through sparse local calibration.** If a landholder can provide one or a few local sensors, how much can a separate calibration layer improve local prediction while preserving independent validation?

This framing is close in spirit to two lines of previous work but differs in its validation emphasis. Process-oriented downscaling studies such as the Equilibrium Moisture from Topography, Vegetation and Soil model (EMT+VS; Ranney et al., 2015) explicitly use fine-resolution topographic, vegetation and soil information to redistribute coarse soil moisture to 10–100 m scales, and show that soil and vegetation information can improve spatial patterns when those inputs are sufficiently detailed. Recent machine-learning tools such as `mlhrsm` (Peng et al., 2024) instead emphasise application-ready high-resolution mapping, uncertainty estimation and spatial–temporal analysis from in-situ sensors, satellite products and land-surface covariates. The present report sits between these traditions: it compares a statistical and a process-based 30 m product using the same model-agnostic dense validation table, and treats seasonal bias, wet/dry compression and terrain-stratified error as primary validation targets rather than secondary diagnostics.

2 Relationship to the companion model paper

The purpose of this report is not to re-rank every model in the development ladder. Instead, it takes the two final comparison models from the companion paper and evaluates them under a different validation geometry: dense within-site observations. Table 1 summarises the division of evidence.

Table 1: How this dense validation report complements the companion model paper.

Companion model paper	Dense validation report
Develops the full model ladder, models 1–8.	Uses the final comparison models only: model 6 RF and model 8 process.
Evaluates transfer across OzNet stations, spatial blocks and years. Primary question: which validation design estimates national transfer?	Evaluates transfer to independent dense point datasets at paddock scale. Primary question: does the 30 m field preserve local spatial patterns and bias structure?
Focuses on regional/station-level skill, residual decomposition and product generation. Uses station-network evidence to justify final model choices.	Focuses on seasonal/wetness bias, terrain-conditioned error, spatial clustering and sparse local adaptation. Uses dense-site evidence to diagnose deployment reliability and local calibration needs.

3 Dense validation datasets

3.1 Sites and validation roles

The analysis uses three dense datasets, each contributing a different kind of evidence (Table 2). Esdale provides the strongest modern within-site terrain coverage but only autumn–winter dates. Tarrawarra provides unusually dense spatial sampling through a full campaign year; because the raw TDR points are denser than the model grid, Tarrawarra is evaluated after aggregation to model grid-cell support. model 6 RF also has a known **SMIPS TotalBucket-zero** caveat for this historical period and should be read partly as a missing-coarse-anchor stress test. Llara provides the strongest seasonal/temporal validation because the profile-mean probe record spans multiple years, but its point density is lower and the two paddocks should be interpreted separately.

Table 2: Dense validation site inventory. Values shown here reflect the current analysis cache and should be checked against the final model-agnostic tables before submission.

Site	Main role	Supports	Dates	Period	Seasons
Esdale	Spatial/terrain microscope	79	9	2025-04-30–2025-07-17	Autumn, win
Tarrawarra	Extreme spatial-density stress test	178	19	1995-09-25–1996-11-29	Spring, summ
Llara	Seasonal/temporal microscope	32	955	2021-10-19–2024-06-30	Spring, summ

3.2 Observation harmonisation

Each dense validation dataset measures soil moisture using methods and depth supports that differ from each other and from the training datasets used for the original model development. The validation target is therefore treated as volumetric soil moisture percentage at the best available site-specific support, not as a perfectly interchangeable sensor quantity.

Llara measurements are obtained from continuously recording in-situ probes that measure through a profile to approximately 1.60 m depth. The validation uses the profile-mean depth from these sensors so that each timestamp contributes a single local soil-moisture estimate. Esdale measurements are handheld TDR probe readings to approximately 10 cm depth. Tarrawarra measurements are TDR probe readings to approximately 15 cm depth.

Tarrawarra also requires explicit spatial-support harmonisation. The campaign sampling is much denser than the model prediction grid, so scoring raw probe observations would over-weight sub-pixel variability that neither a 30 m statistical model nor a 30 m process model can resolve. For Tarrawarra, validation is therefore performed after aggregating observations to the model support scale: within each model/date/grid-cell combination, observed TDR soil moisture is averaged and the prediction is sampled once from the corresponding raster cell. Residual sub-pixel variation is treated as an observational/support mismatch rather than direct model error. This provides an additional diagnostic: grid cells with high within-cell TDR variance mark locations where no 30 m model should be expected to score perfectly unless it explicitly represents sub-pixel structure.

Dense validation site terrain context

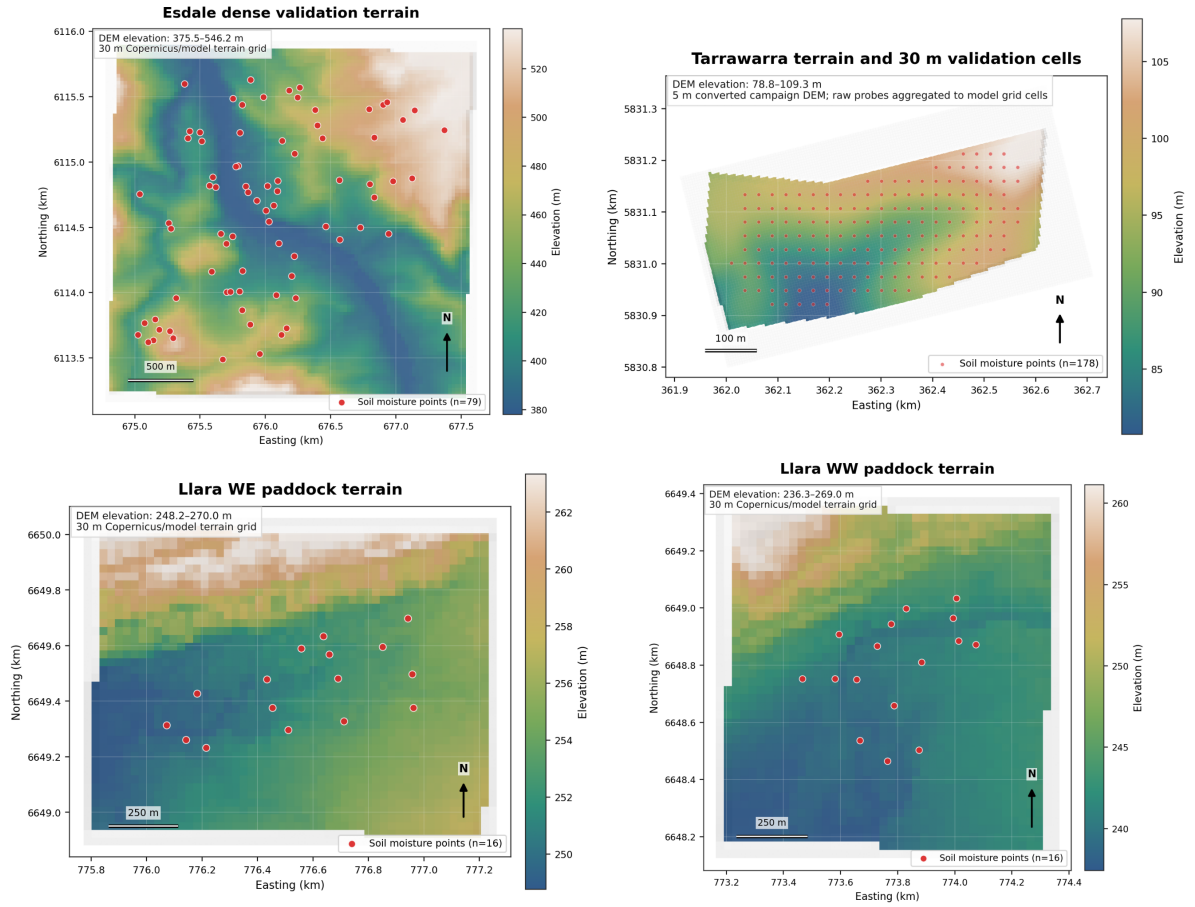


Figure 1: Dense validation site terrain context. Esdale, Llara WE and Llara WW use the 30 m Copernicus/model terrain grid; Tarrawarra uses the converted 5 m campaign DEM. Red points are soil-moisture validation supports: probe locations for Esdale and Llara, and aggregated model-grid-cell centroids for Tarrawarra.

3.3 Model predictions and gridded products

For each site and observation date, model predictions are generated or sampled from cached gridded products and extracted at the relevant observation support: probe coordinates for Esdale and Llara, and model grid-cell supports for Tarrawarra after aggregation. The validation protocol is deliberately model agnostic: after prediction, each model is represented only by a table of support/date predictions and associated model-input strata. This allows model 6 RF and model 8 process to be compared without relying on either model’s internal representation.

[Caution: For Tarrawarra, the current model 6 RF prediction table has a **SMIPS TotalBucket-zero caveat** because historical SMIPS coverage was unavailable or returned zero-valued inputs for the model run. Do not interpret Tarrawarra model 6 RF skill as a normal operational model6 result until this is resolved or explicitly framed as a coarse-anchor ablation.]

4 Model-agnostic validation protocol

4.1 Common prediction table

All validation analyses begin from a common table with one row per model–support–date combination:

Field group	Required / derived columns
Identity	<code>site</code> , <code>model_name</code> , <code>point_id</code> or grid-cell identifier, <code>date</code>
Location	<code>lon</code> , <code>lat</code> , optional projected coordinates
Target and prediction	<code>obs_sm_pct</code> , <code>pred_sm_pct</code> , <code>residual = pred - obs</code>
Temporal strata	season, year, date block, wetness state
Static model inputs	elevation, slope, TWI, HLI, accumulation, aspect components, SLGA soil properties
Dynamic model inputs	SMIPS summaries where available, antecedent rainfall, PET and VPD summaries

4.2 Metrics

The metrics match the companion paper so that results are comparable:

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2} \sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}}, \quad (1)$$

$$\text{NSE} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}, \quad (3)$$

$$\text{bias} = \frac{1}{n} \sum_i (\hat{y}_i - y_i), \quad (4)$$

$$\text{ubRMSE} = \sqrt{\max(\text{RMSE}^2 - \text{bias}^2, 0)}. \quad (5)$$

Here y_i is observed soil moisture and $\hat{y}_i$ is predicted soil moisture. We report pooled metrics by site and model, seasonal metrics, per-support metrics and paired model differences at matched support/date rows. Pearson r measures whether predictions track temporal or spatial variation, whereas NSE penalises both variance errors and mean-level offsets. Where observed variance is low, NSE can be unstable or harsh; interpretation therefore uses Pearson r , RMSE, ubRMSE and bias alongside NSE.

4.3 Blocking and separation of validation from calibration

Stage 1 uses all dense observations only as external validation. Stage 2 uses controlled local calibration subsets but validates on held-out points and dates. The strict spatial+temporal block is treated as the meaningful transfer estimate because it tests whether calibration learned a transferable local relationship rather than memorising a point or date.

5 Stage 1: independent dense-point validation

5.1 Headline site-level skill

The first analysis pools all available support/date rows by site and model (Fig. 2). This figure should be interpreted as an external paddock-scale validation summary, not as a replacement for the OzNet cross-validation ladder in the companion paper.



Figure 2: Stage 1 independent dense-point validation: pooled NSE, Pearson r , RMSE, ubRMSE and bias by site and model. The five metrics are reported together because correlation, efficiency, total error magnitude, bias-corrected error and mean-level offset answer different validation questions.

5.2 Paired comparison of model6 and model8

Because model 6 RF and model 8 process are sampled at the same support/date rows, their residuals can be compared pairwise. The main paired comparison reports $\text{RMSE}(\text{model 6 RF}) - \text{RMSE}(\text{model 8 process})$ and $\text{bias}(\text{model 6 RF}) - \text{bias}(\text{model 8 process})$ on matched supports, with bootstrap uncertainty added in the supplement where appropriate. Positive RMSE differences indicate lower RMSE for model 8 process; positive bias differences indicate that model 6 RF is wetter, or less dry, than model 8 process on the same matched rows.

Planned paired-comparison figure

Create a compact paired model figure with one row per site. Suggested panels: $\text{RMSE}(\text{model 6 RF}) - \text{RMSE}(\text{model 8 process})$, $\text{bias}(\text{model 6 RF}) - \text{bias}(\text{model 8 process})$, and optionally bootstrap intervals for these paired residual summaries. Source table: `outputs/unified_dense_validation/stage1_independent_validation/paired_model_comparison_over`

5.3 Predicted and observed time series

Spatially averaged time series provide a diagnostic of seasonal tracking and mean-level bias. They are not a substitute for support-level scoring, but they make temporal drift and wet/dry compression visually apparent.

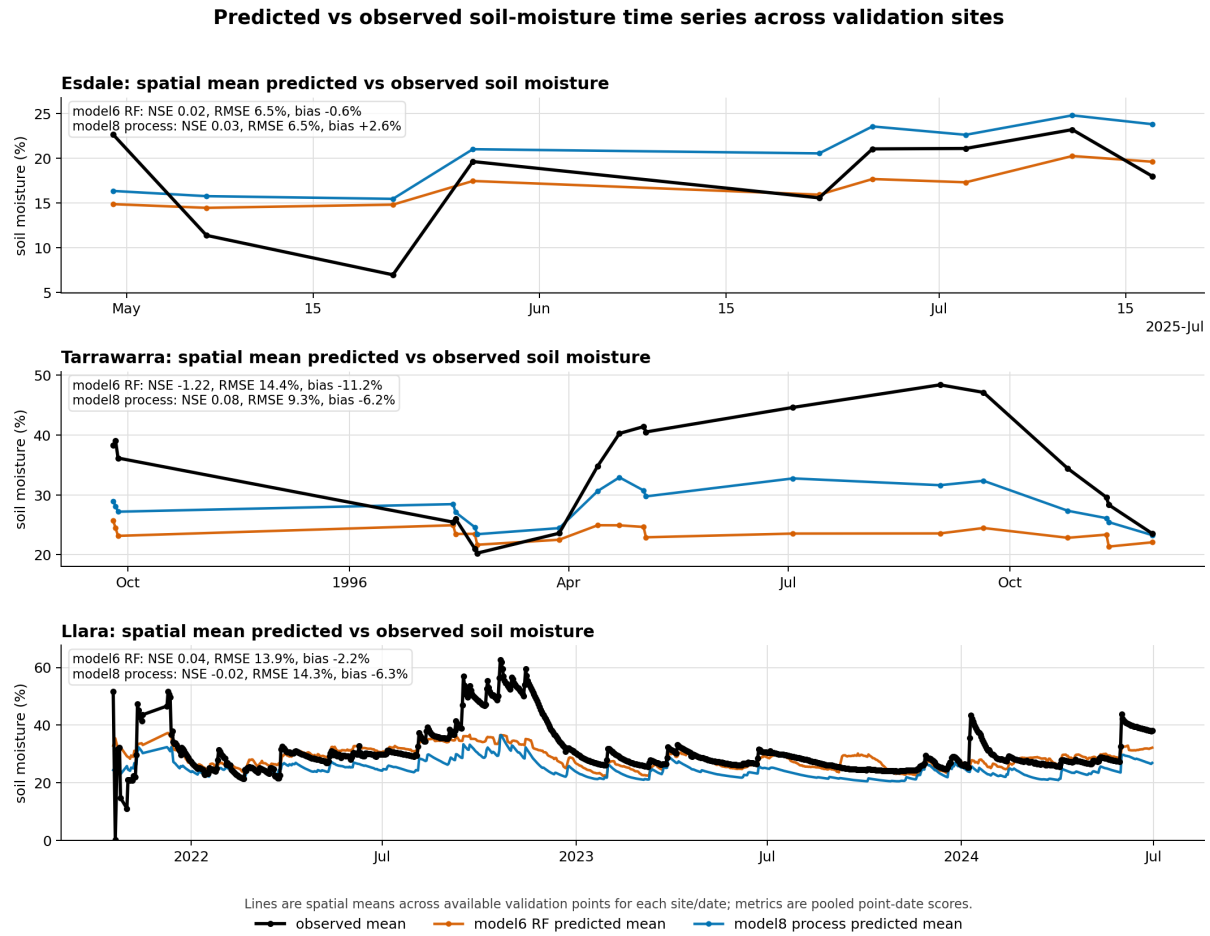


Figure 3: Observed and predicted soil-moisture time series by site. Values are spatial means over available validation points for each date. [TODO: Check whether Llara should be shown as WE/WW paddocks separately in the final manuscript.]

Table 3: Headline independent validation metrics from the current unified analysis cache. For Tarrawarra, n is the number of model/date/grid-cell supports after aggregation, not the number of raw TDR observations.

Site	Model	n	NSE	r	RMSE	ubRMSE	Bias / interpretation
Esdale	model 6 RF	560	0.023	0.238	6.477	6.453	−0.558; weak spatial skill, small mean dry bias.
Esdale	model 8 process	560	0.031	0.462	6.452	5.884	2.645; higher association but wet-biased.
Tarrawarra	model 6 RF	2154	−1.219	0.398	14.402	9.093	−11.169; strong dry bias, interpreted with the SMIPS TotalBucket-zero caveat.
Tarrawarra	model 8 process	2154	0.077	0.853	9.291	6.958	−6.157; strongest Tarrawarra association after support aggregation.
Llara	model 6 RF	29247	0.036	0.267	13.927	13.744	−2.249; slightly better error/bias than model 8 process overall.
Llara	model 8 process	29247	−0.017	0.453	14.306	12.855	−6.278; stronger association but larger dry bias.

5.4 Seasonal and wetness-state bias

The dense validation should emphasise seasonal and wetness-state bias because a model can have acceptable pooled error while systematically overpredicting dry states or underpredicting wet states. This is particularly relevant to trafficability, drought stress and paddock management decisions.

5.5 Terrain and model-input stratification of error

Dense validation allows error to be stratified by the same model inputs that drive the downscaling field, including elevation, slope, TWI, HLI, accumulation, aspect components, SLGA soil properties and antecedent meteorological summaries. The main text should show only the most notable strata; complete strata tables belong in the supplement.

Planned terrain-stratification figure

Recommended final figure: rows = site, columns = model, colour = bias or RMSE contrast between low/mid/high strata. Show only the top 2–3 variables per site/model ranked by combined bias and RMSE range. Source tables: `metrics_by_terrain_strata.csv` and `notable_terrain_error_strata.csv`.

[TODO: Write one paragraph per site: Esdale terrain signal, Tarrawarra terrain/spatial signal, Llara seasonal/field signal. Avoid overinterpreting variables that are collinear or dynamic strata masquerading as terrain.]

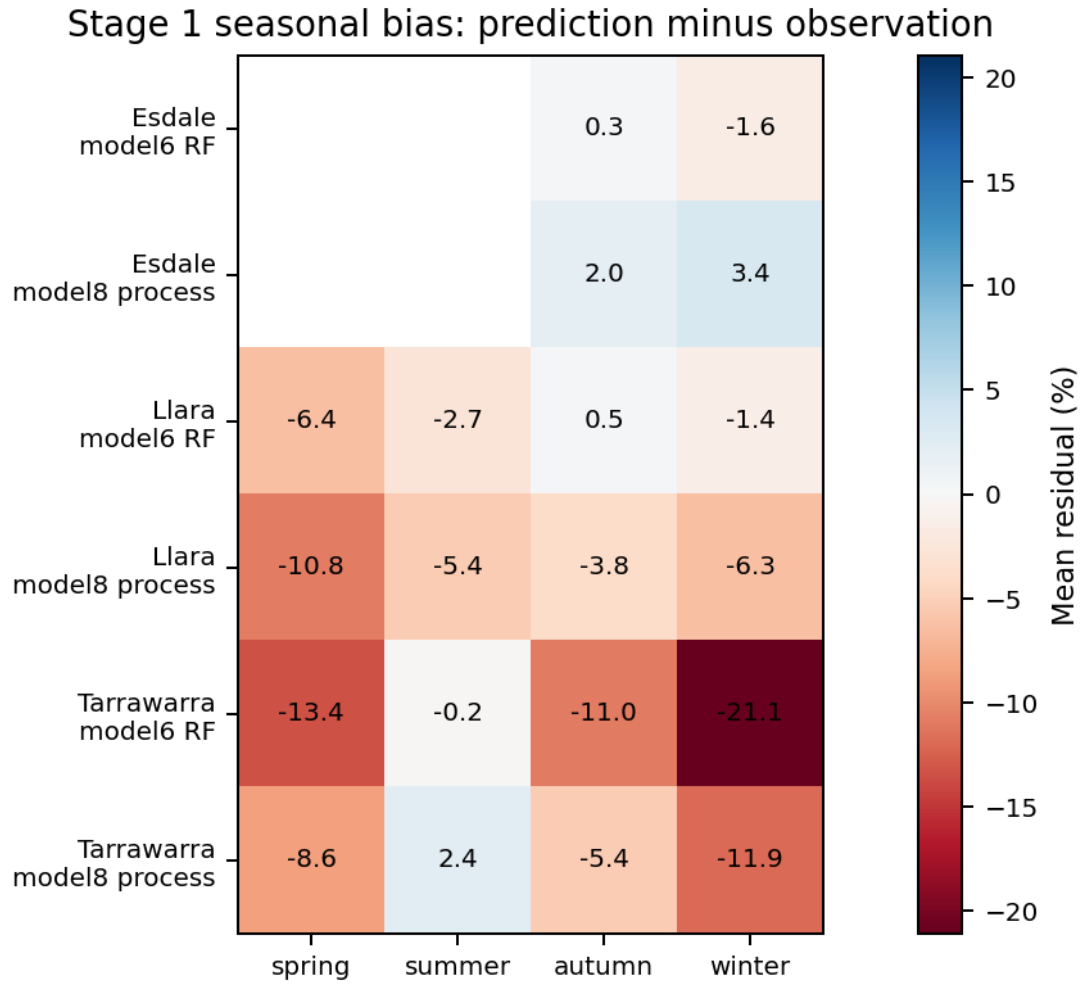


Figure 4: Seasonal bias by site and model. Residuals are prediction minus observation; positive values indicate overprediction. [TODO: In final caption, name the largest seasonal biases and note which site has insufficient seasonal coverage.]

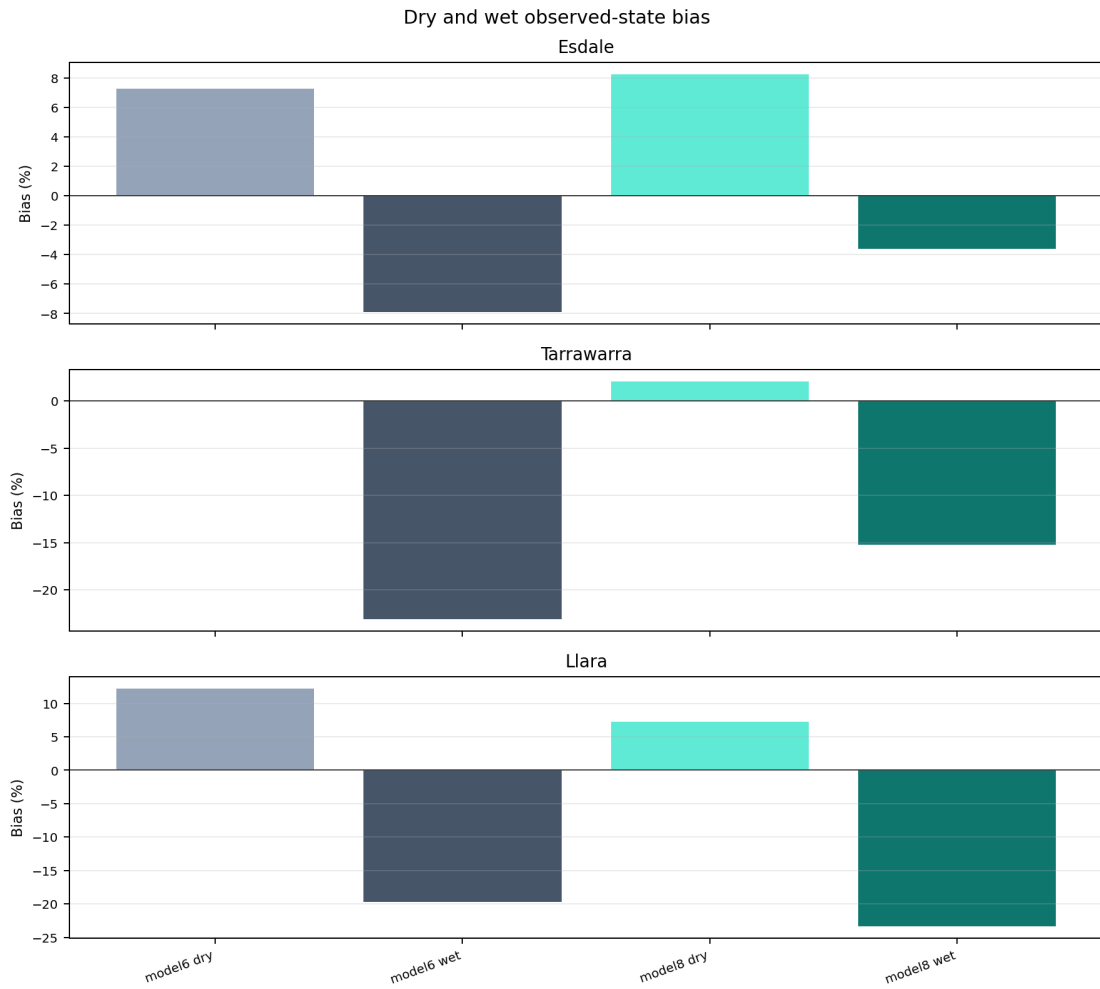


Figure 5: Dry/wet observed-state bias. The driest and wettest observed quartiles test whether models compress extremes. [TODO: Add final interpretation: overprediction in dry quartile, underprediction in wet quartile, or site/model exceptions.]

5.6 Spatial clustering of prediction quality

Prediction-quality maps test whether poor performance is spatially organised. For publication, show support-level maps of NSE, Pearson r , RMSE, ubRMSE and bias where sample size permits, plus a paired model-advantage map where useful. Interpolated quality surfaces can aid visual diagnosis but should be labelled as diagnostic interpolation, not model output.

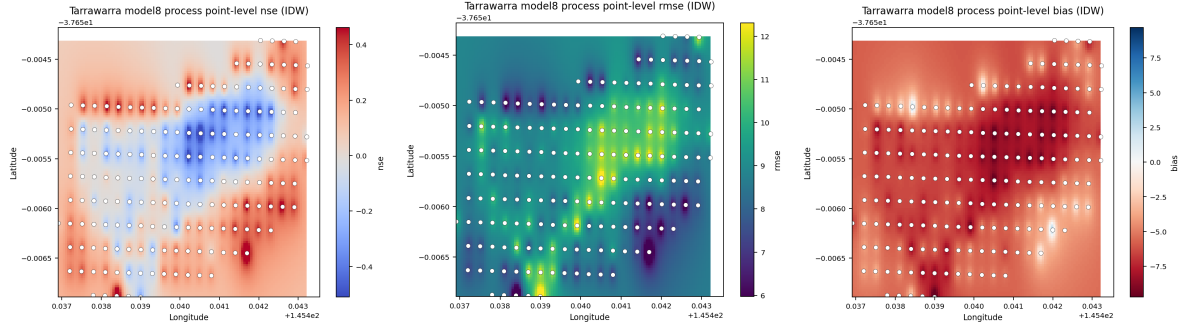


Figure 6: Example spatial prediction-quality surfaces for Tarrawarra model 8 process. These are diagnostic interpolations of validation metrics at aggregated model-grid-cell support, not gridded model predictions. [TODO: Replace or extend with a final multi-site figure that uses consistent colour scales and includes paired model-advantage maps.]

5.7 Gridded prediction examples

The dense validation ultimately evaluates gridded predictions. Representative dry and wet dates should therefore be shown using actual model rasters rather than interpolated point surfaces.

6 Stage 2: sparse local calibration and local spiking

6.1 Rationale

Independent validation tells us where the national/global model fails. A deployment question follows: can a small amount of local information correct site-level or terrain-conditioned bias without requiring dense sampling at every property? This stage should be interpreted as a local adaptation experiment, not as independent validation of the base models.

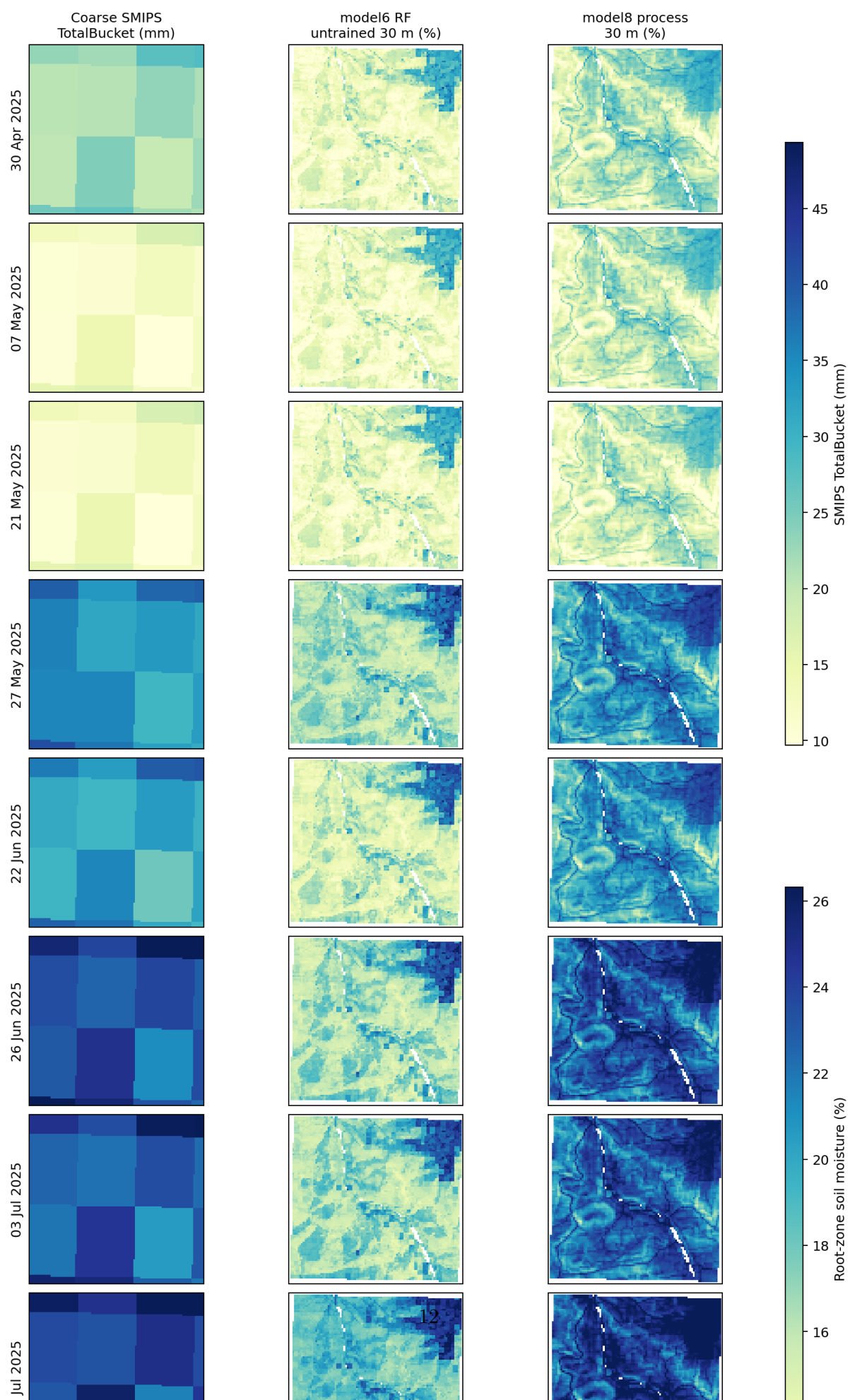
The experiment should mimic realistic local information. A new landholder will usually not know where the model performs badly, but may know where the landscape is generally wet, dry, low, high, clayey or exposed. Calibration point selection should therefore compare random selection against simple landscape-knowledge proxies and model-informed extremes.

6.2 Calibration layers and point-selection strategies

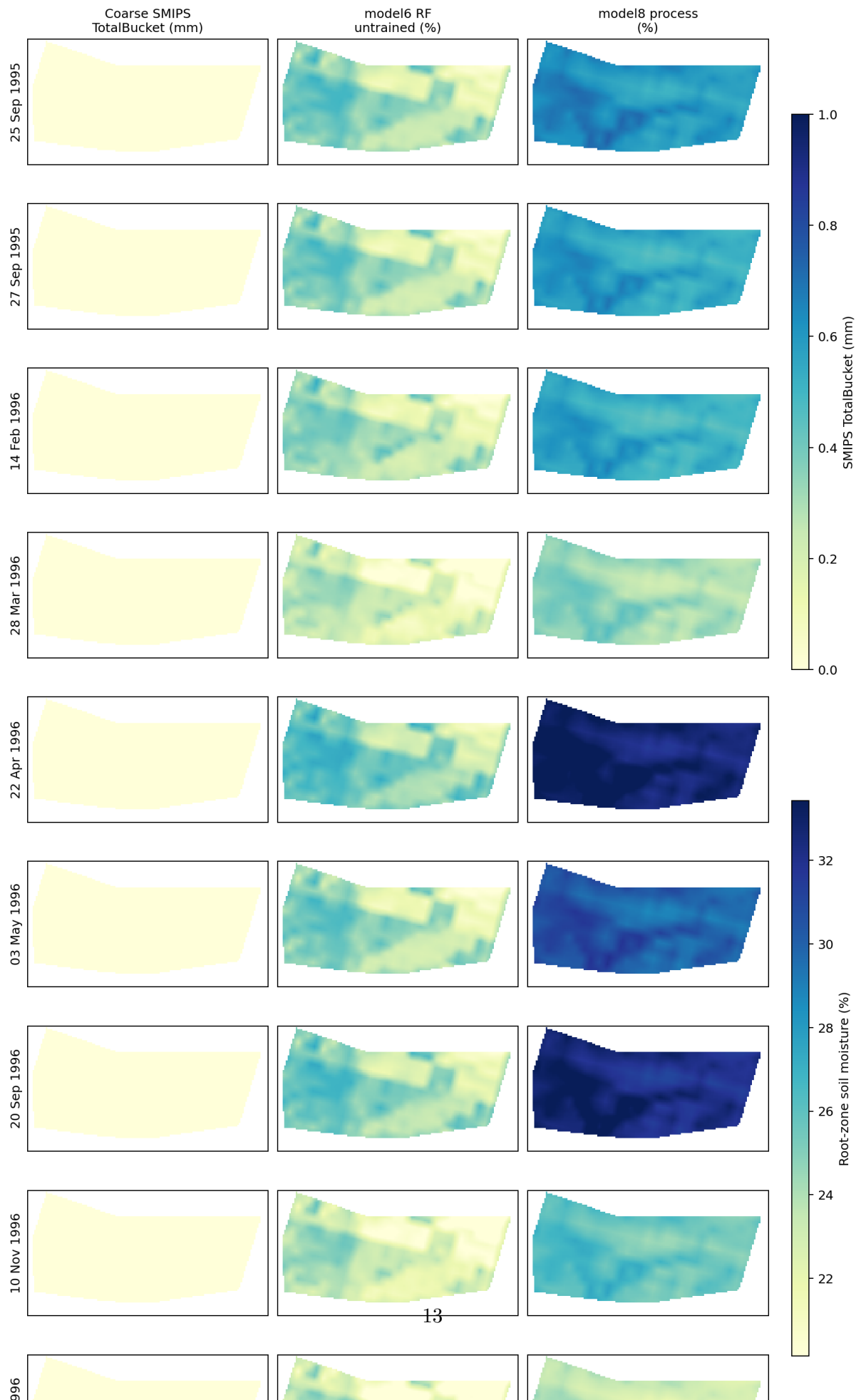
The current calibration layers are:

1. **Bias offset:** one constant residual correction.
2. **Seasonal offset:** separate residual offsets by season where enough dates exist.
3. **Affine correction:** local linear rescaling of prediction level.
4. **Residual ridge layer:** residual model using selected model-input covariates.

Dense campaign soil-moisture maps: coarse estimate vs model6 RF vs model8 process



Tarrawarra campaign maps in the converted 5 m DEM footprint



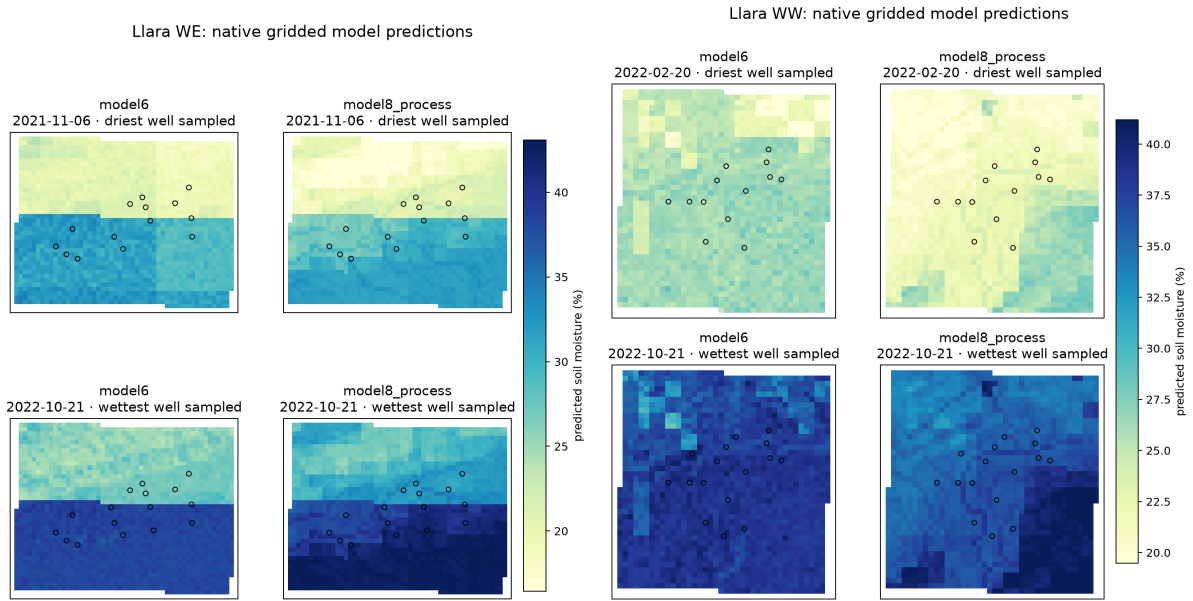


Figure 9: Llara gridded prediction examples for the WE and WW paddocks, shown separately. [TODO: Consider separate full-width panels in final layout if the two-paddock figure is too dense.]

Point budgets are one, three, five, ten, 25 %, 50 % and all eligible local calibration points. Validation is performed on held-out points and dates, with the strict spatial+temporal block treated as the primary transfer estimate.

[TODO: State exactly which selection strategies are retained in the final protocol. Candidate final set: random, field-knowledge wet/dry proxy, global-prediction extremes, terrain-representative.]

6.3 Learning curves

6.4 One-sensor placement strategies

6.5 Do process and statistical models respond differently?

The process-vs-statistical comparison is both technical and philosophical. A statistical downscaler may already encode nonlinear terrain and soil relationships learned from the station network, but it may be brittle outside the training distribution. A process model may have simpler temporal dynamics and fewer degrees of freedom, but a local layer may correct its site-level state more cleanly. Dense datasets allow this difference to be measured as the marginal gain from the same sparse local information.

7 Discussion

7.1 What dense validation adds beyond station validation

[TODO: Synthesis paragraph. Suggested claim: the companion paper establishes regional/station transfer; dense validation reveals whether the 30 m product has meaningful within-site spatial structure and whether residuals organise by terrain, season and wetness state.]

7.2 Model complementarity at paddock scale

[TODO: Discuss whether model 6 RF and model 8 process fail in the same places. Use paired comparisons and maps. Avoid a single global winner unless the paired evidence is strong and

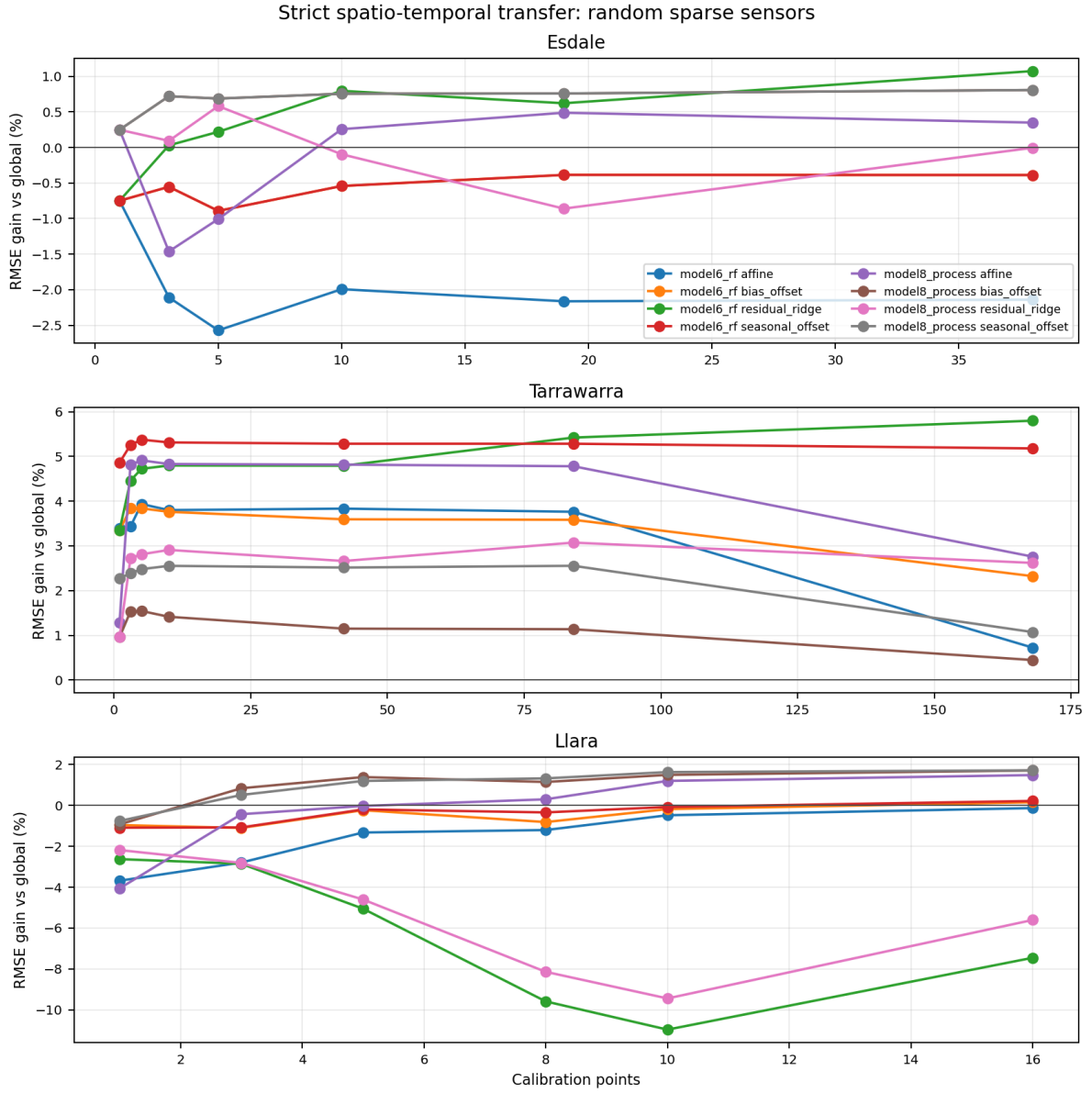


Figure 10: Sparse local-calibration learning curves under strict spatial+temporal blocking. Positive values indicate lower RMSE after calibration. [TODO: Update caption with the minimum point budget needed to achieve material improvement by site and model.]

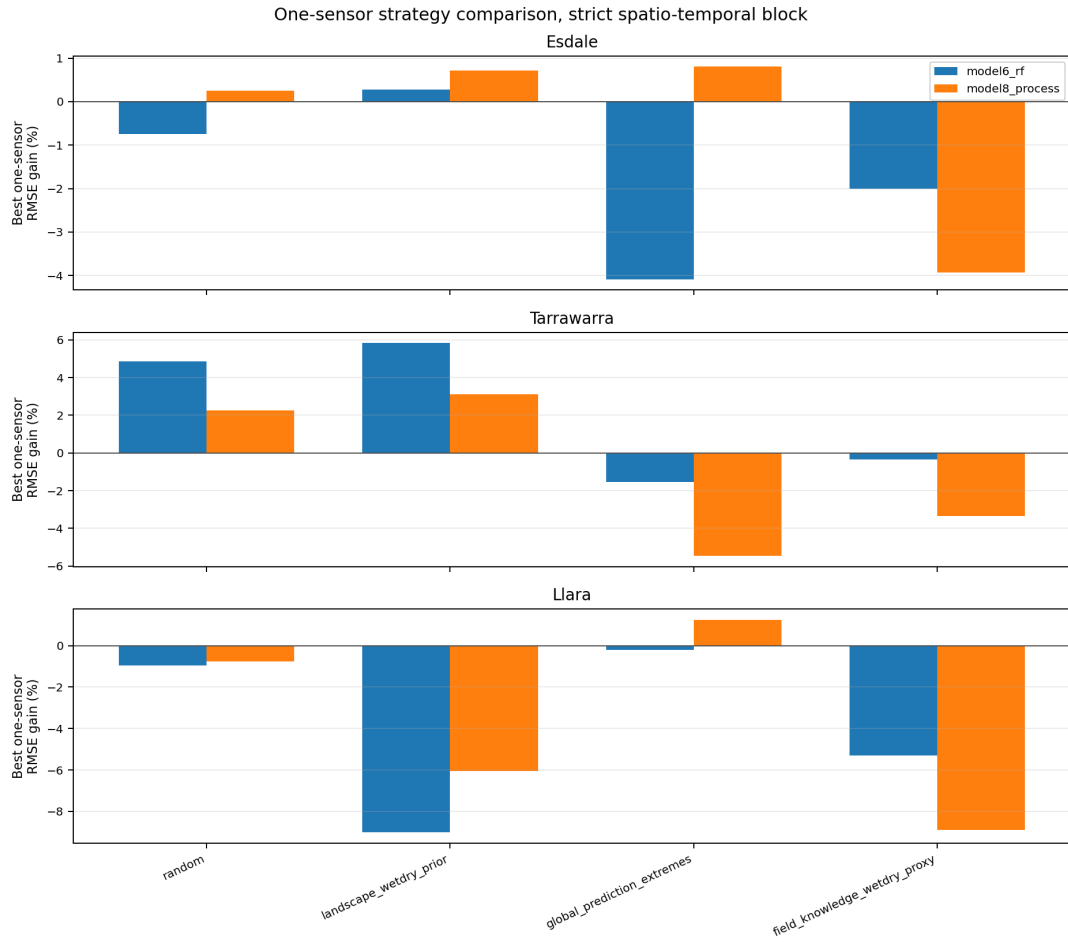


Figure 11: One-sensor selection strategy comparison. This figure addresses the deployment question: if only one local point can be instrumented, should it be chosen randomly, by landscape knowledge, or by model-informed wet/dry extremes? [TODO: Add final inference by site.]

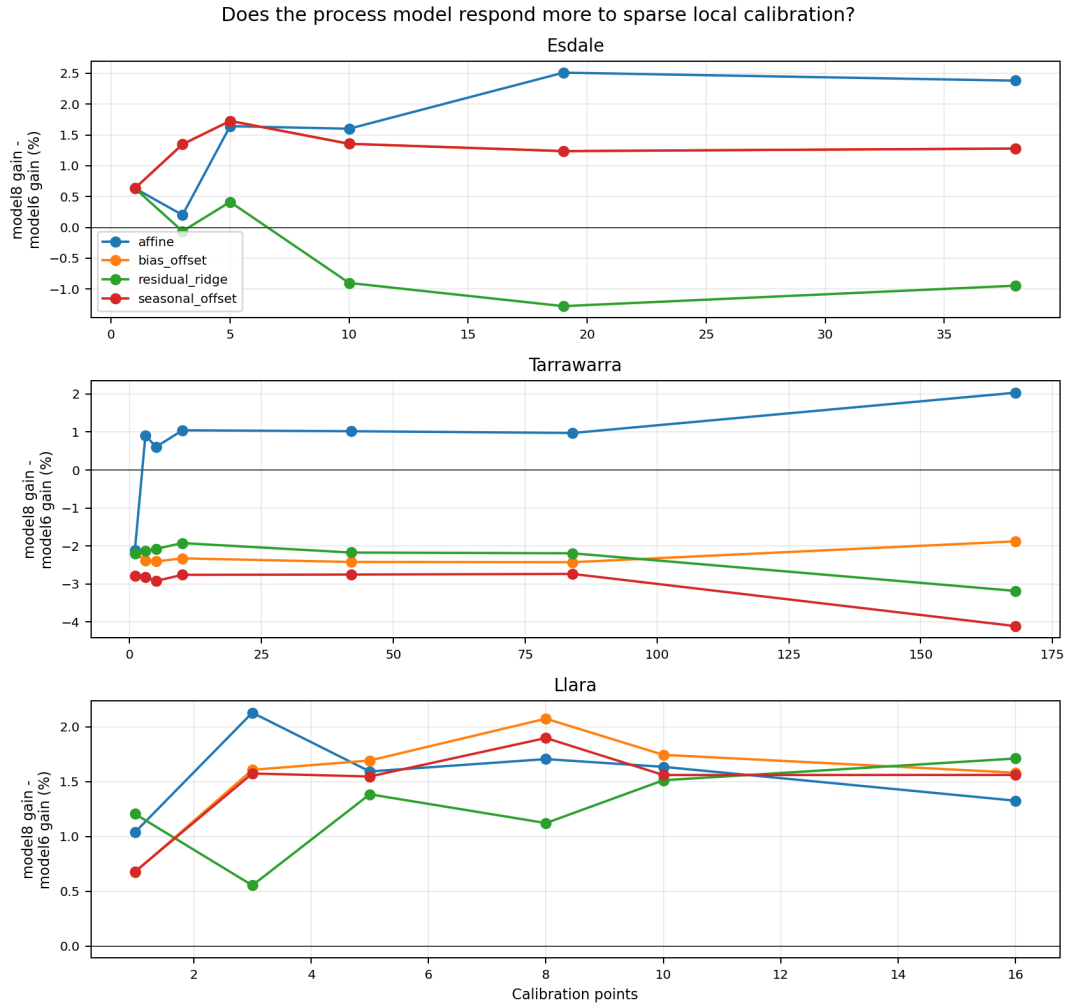


Figure 12: Process-vs-statistical local calibration responsiveness. Positive values indicate that model 8 process gained more RMSE improvement than model 6 RF from the same sparse local information budget. [TODO: Confirm whether the final figure should use random selection only or the best realistic selection strategy per site.]

caveats are resolved.]

7.3 Implications for local deployment

[TODO: Discuss the one-sensor / few-sensor deployment story. Frame local calibration as a separate adaptation layer placed on top of a national/global model, not as leakage into independent validation.]

7.4 Future public training and validation data

[TODO: Discuss likely value of additional public dense datasets, sensor networks, campaign data and remote/proximal sensing products. Separate data that improve model training from data that improve independent validation.]

8 Limitations

- Esdale has strong spatial/terrain coverage but short autumn–winter coverage.
- Llara has strong temporal/seasonal coverage but only 32 probes across two paddocks.
- Tarrawarra has exceptional raw point density; validation now aggregates those points to model grid cells by date before scoring. The current model 6 RF SMIPS TotalBucket-zero caveat prevents direct interpretation as an operational model 6 RF result. High within-cell TDR variance should be treated as support mismatch unless future models explicitly represent sub-pixel structure.
- Soil-moisture depth definitions and profile integration differ across datasets and must be documented carefully.
- Dense datasets remain limited in climate and soil coverage relative to the intended national product.
- NSE can be unstable when observed variance is small; Pearson r , RMSE, ubRMSE and bias must be interpreted alongside it.
- Local calibration experiments are deployment sensitivity analyses, not independent validation of the base models.

9 Conclusions

[TODO: Write as 4–6 numbered conclusions. Suggested structure: (1) dense validation is a distinct evidence layer; (2) headline independent skill and caveats; (3) seasonal/wetness/terrain bias; (4) model6/model8 complementarity; (5) sparse local calibration implications; (6) priority next data.]

Code and data availability

All validation code and report-generation scripts are maintained in the `DMM_validation` repository. Model definitions and the companion paper are maintained in the `DownscalingMoistureModel` repository. [TODO: Add repository URLs, branch names, commit hashes, data access statements and any restrictions before submission.]

Supplementary material plan

- Full model-agnostic prediction tables for Esdale, Tarrawarra and Llara.
- Full per-site/model metric tables.
- Full terrain/model-input strata tables.
- Full paired support/date model comparison tables.
- Local calibration design tables, selected calibration points and replicate-level learning-curve outputs.
- Additional prediction-quality maps and gridded dry/wet map examples.

References

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